

Simona Kocour

PhD Student in Computer Science, ELLIS PhD Program

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Research Interests

3D Scene Representations, Dynamic Scene Understanding, Vision-Language Models, Neural Rendering, World Models, and Foundation Models.

Publications

Structured 3D Scene Representations Improve Spatial Reasoning Across Benchmarks

Simona Kocour, Yutong Chen, Torsten Sattler, Sergey Prokudin, Siyu Tang

Under review

Remove360: Benchmarking Residuals After Object Removal in 3D Gaussian Splatting

Simona Kocour, Assia Benbihi, Torsten Sattler

In Proceedings of the European Conference on Computer Vision (ECCV) Workshops, Structure-from-Motion in the Age of Deep Learning (SFM-DL), 2026. [ [Code](#)]

Research & Work Experience

Visiting PhD Student, ETH Zurich

Aug 2025 – Aug 2026

- Developed structured 3D scene representations that improve spatial reasoning capabilities of foundation Vision-Language Models.
- Research on spatial intelligence, scene graphs, and semantic scene understanding in 3D.
- Working with Prof. Siyu Tang and Dr. Sergey Prokudin.

PhD Student, Czech Technical University

Sept 2024 – Present

- Research on neural scene representations, semantic understanding, and controllable 3D scene editing.
- Designed and benchmarked new evaluation protocols analysing residuals left after object removal in 3D scenes.
- Selected as a member of the ELLIS PhD program.

Machine Learning Researcher, Emplifi

Nov 2020 – Mar 2024

- Developed NLP pipelines for large-scale social media analysis.
- Fine-tuned transformer architectures (BERT) for classification and summarization.
- Built large-scale data processing pipelines using PySpark, AWS and Databricks.

Data Scientist, Aures Holdings

Aug 2019 – Oct 2020

- Designed predictive pricing algorithms using robust regression frameworks, directly impacting company pricing strategies.
- Deployed machine learning models to predict customer purchasing behavior.

Education

PhD in Computer Science	ELLIS PhD Program Advisors: Torsten Sattler (CTU Prague), Siyu Tang (ETH Zurich)	2024 – Present
MSc in Mathematics and Statistics	Masaryk University, Czech Republic	2017 – 2019
BSc in Mathematics	Masaryk University, Czech Republic	2014 – 2017

Technical Skills

3D Computer Vision:	3D Gaussian Splatting, NeRFs, Neural Rendering, Structure-from-Motion (SfM), 3D Reconstruction, Scene Graphs
Machine Learning:	PyTorch, Transformers, Vision-Language Models (VLMs), Multi-Modal Learning, Open-Vocabulary Segmentation, Deep Learning
Programming:	Python, R, SQL, Git, Docker
Data & Cloud:	AWS, Databricks, PySpark

Languages

Slovak (Native) · **English, Czech** (Full Professional Proficiency) · **German** (Intermediate)